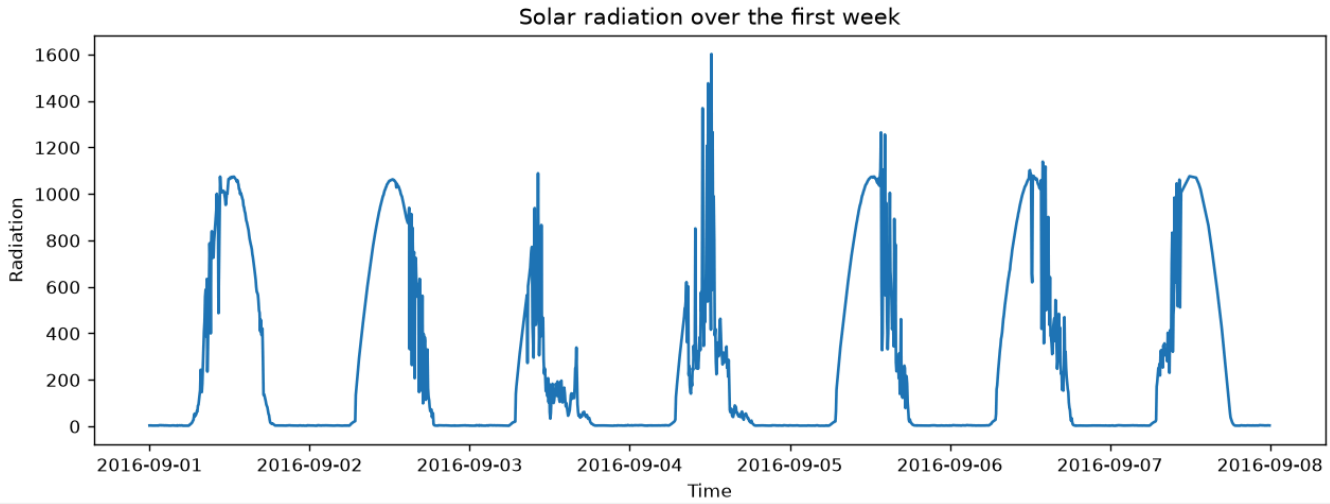
STAC WARS PS2 PART B

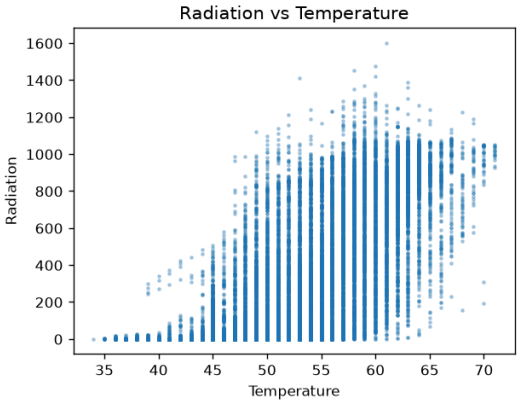
TEAM NAME

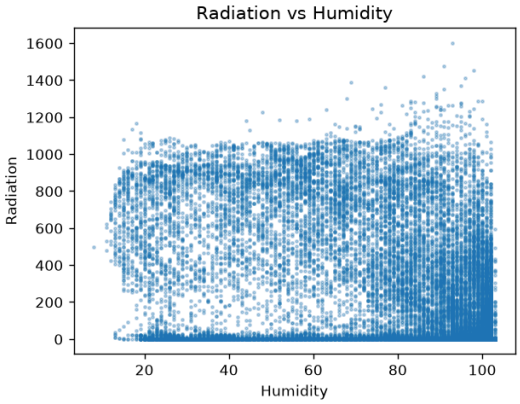
Multiple different graphs were plotted, including:

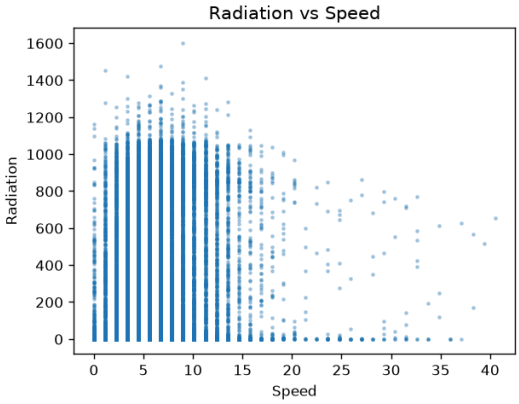
- Radiation over the first week



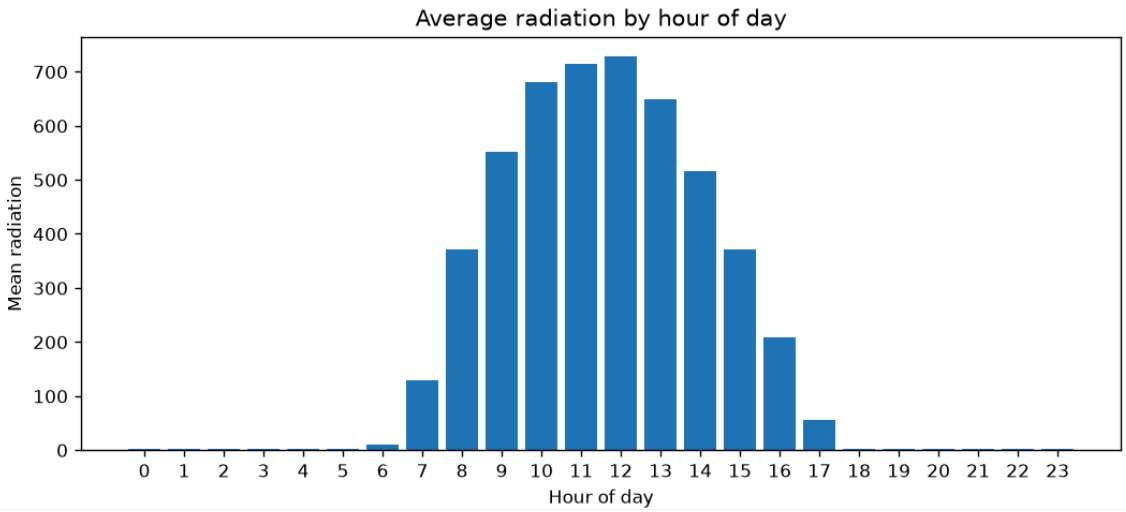
- Scatter plot of Radiation vs Temperature, Humidity and Speed



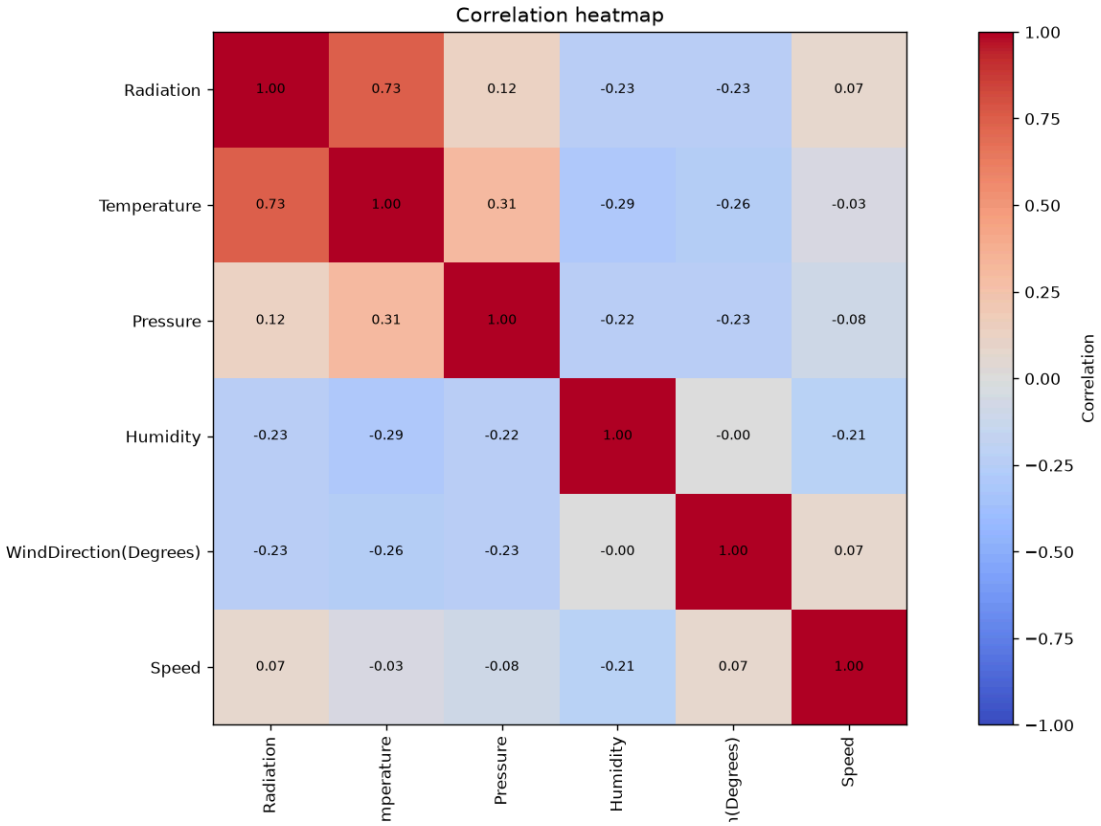




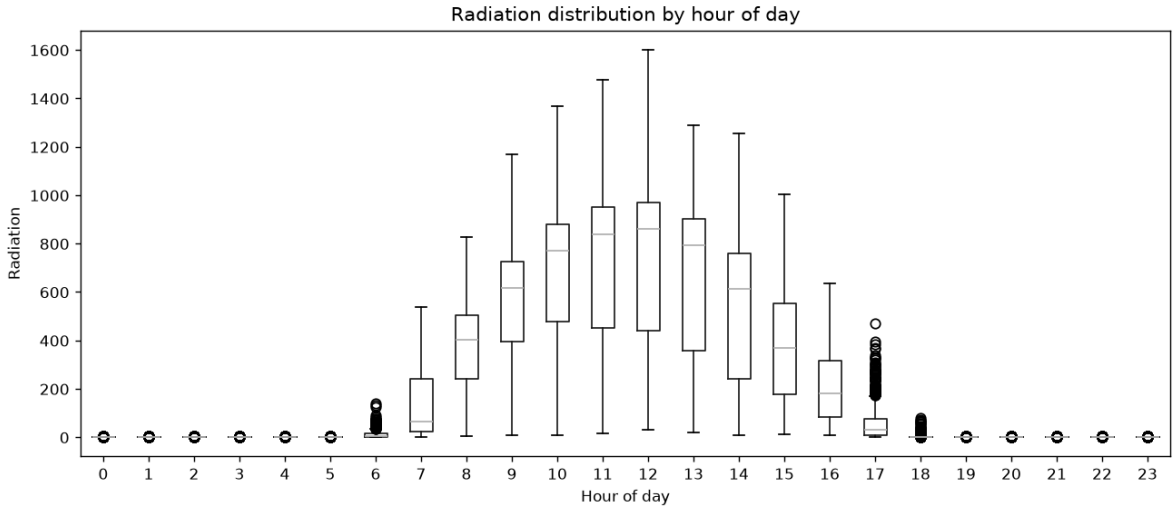
- Bar graph of Mean Radiation by hour of day



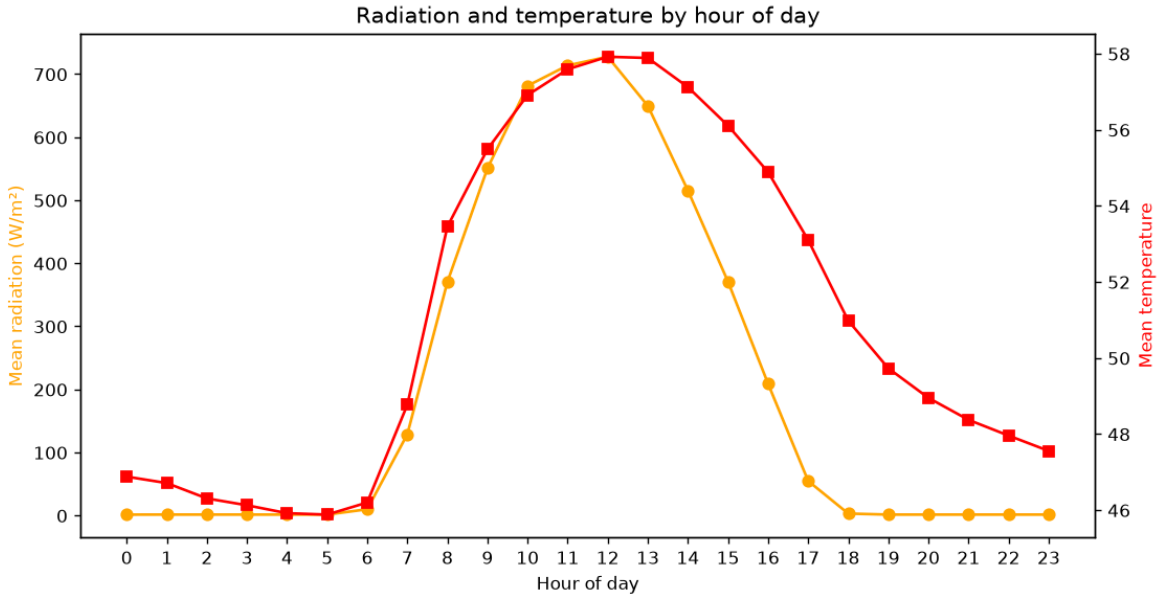
- Correlation Heatmap between all important conditions



- Boxplot of Radiation distribution per hour



- Comparison between Radiation and Temperature throughout the day



SUMMARY OF FINDINGS:

1. **Solar radiation is highest around noon (hour 12), and is close to zero from about 18:00 to 06:00.**
   - Every day forms a bell-shaped bump, which peaks around 12.
   - Average radiation by hour is highest by 12.
   - Radiation begins to rise by 7, and is back to almost zero by hour 18.
   - On the record day, the peak reading occurred at 12:15:04, which is 6 hours 7 minutes after sunrise.
2. **Temperature has the strongest relationship with radiation.**
   - Temperature has a correlation of about +0.73.
   - Humidity, wind direction, pressure and wind speed are all weakly, or not related.
   - Radiation and temperature rise together, while humidity tends to move in the opposite direction to both.
   - In the radiation vs temperature plot, radiation is near zero for temperatures under 45.
   - No clear conclusions could be made from radiation vs wind speed graph, as wind speed rarely exceeded 20, and radiation was even for most wind speeds.
3. **The highest reading in the dataset, 1601.26 W/m² on 2016-09-04, is a clear outlier**
   - Compared with the clear-day peaks of about 1070 W/m², the highest reading is much higher, and is probably linked to a solar flare.
   - In the first week, 02,05,06,07 have smooth graphs, but 03 and 04 have jagged peaks that end up higher than the other days. This could be linked to a solar flare.
4. **Cool finding - Temperature rises together with the sun but falls much slower than radiation.**
   - Both temperature and radiation peak together, around 12.
   - Radiation dips to 0 by 18, while temperature retains throughout the day.
   - This suggests that the air warms up with sunlight, but dissipates slowly.